	Pimpri Chinchwad Education Trust's **Pimpri Chinchwad College of Engineering**	
Assignment No. 4		

Create a Freestyle Jenkins job: Configure a job to print "Hello, Jenkins!" in the console output and view logs.

Objective:
To understand and demonstrate the creation of a Freestyle Jenkins job by executing a basic build command and viewing the console output.

Software / Tools Required

- Jenkins (installed locally or on a server)
- Web browser (Chrome / Firefox)
- Java (JDK 17,21 or above)

Theory

1. Jenkins

Jenkins is an open-source automation tool built in Java and primarily used for Continuous Integration (CI) and Continuous Delivery/Deployment (CD) in software development. It is used to automate the build and deployment of an application continuously each time modifications are made to the source codes. Jenkins is able to perform its tasks through integration with version control tools like Git. The tool is able to automate different tasks through plugins.

2. Types of Jobs in Jenkins (Add theory)

Jenkins offers different job types to automate different activities executed in the software development life cycle.

The first is a Freestyle Job, which is the easiest job to configure. It is recommended to begin the journey with a freestyle job.

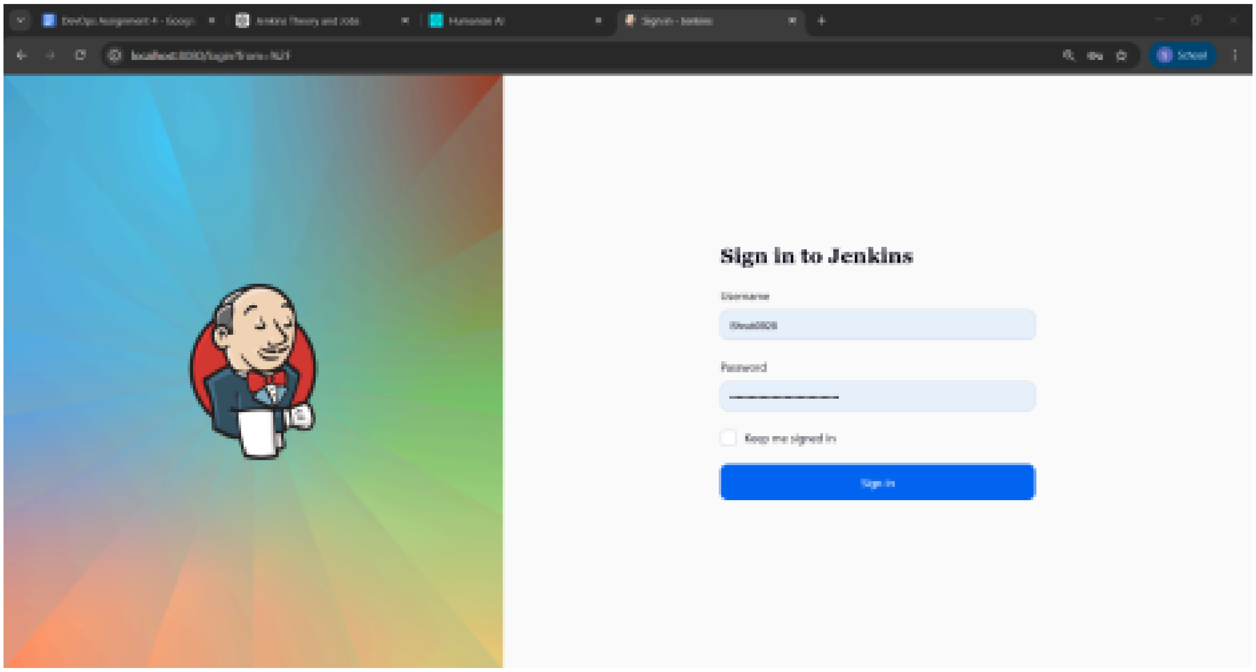
The second is a Pipeline Job, which is primarily used to implement the build, test, and deploy process for the application.
A multibranch pipeline job is created to automatically create a pipeline for a given repository. A folder job is mainly used to manage different jobs in a folder. These jobs are very important for efficiently handling small to big projects.

3. Jenkins Configuration. (add theory)

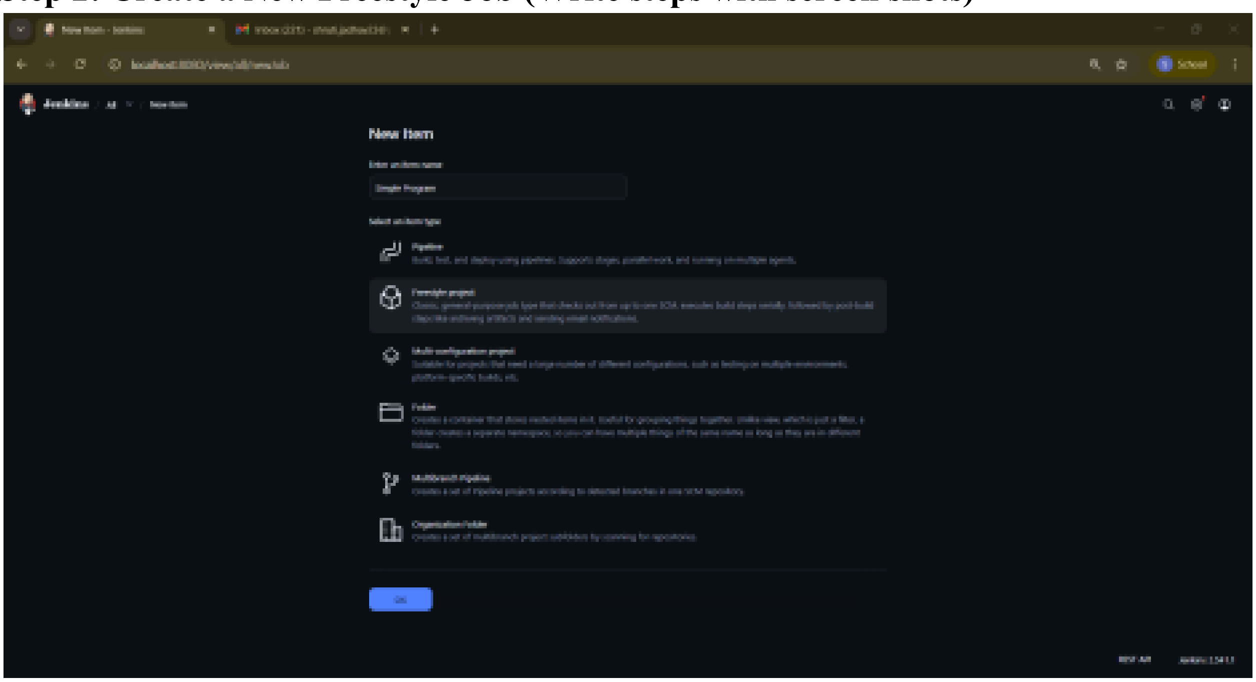
Jenkins configuration involves the configuration of settings, tools, or plugins that Jenkins requires in order to function adequately. Jenkins configuration includes the setup of global settings such as Java, build tools, security settings, user permissions, and other Jenkins plugins. Jenkins configuration additionally involves managing Jenkins plugins, which help in extending Jenkins, and the setup of variables. An adequately configured Jenkins is essential in order for automation, as well as execution of processes, to be carried out adequately.

Procedure

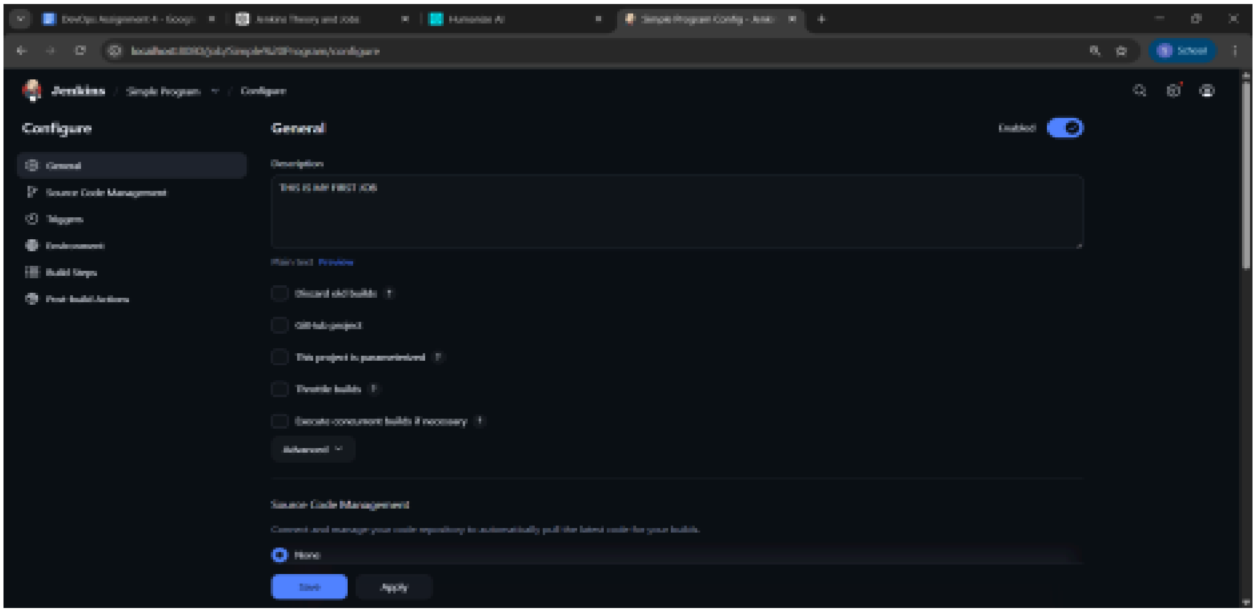
Step 1: Login with Jenkins credentials

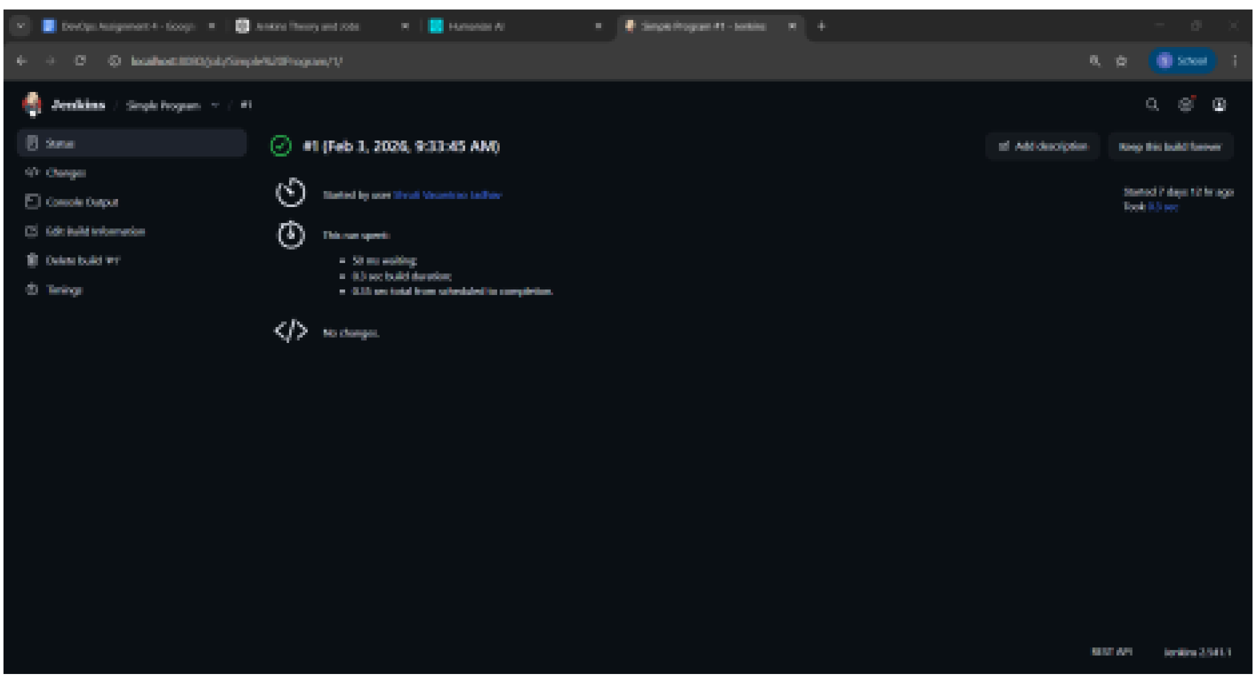


Step 2: Create a New Freestyle Job (Write steps with screen shots)

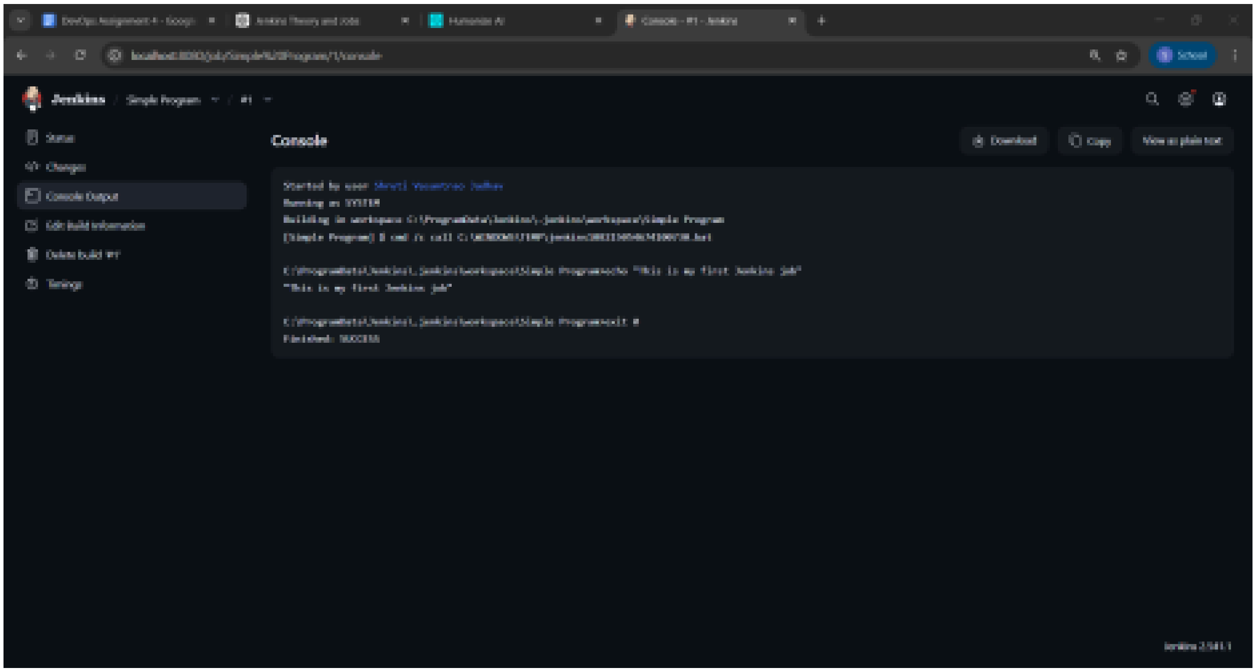


Step 3: Configure the Job (Write steps with screenshot)





Step 4: View Console Output (Screenshot)



FAQ:

1. Why is Jenkins used in DevOps?

Jenkins is used to automate the practice of software development, deployment, and testing within DevOps, and it simplifies the practice of Continuous Integration/Delivery (CI/CD) for developers by automating the triggering of tasks whenever code is modified.

It increases efficiency, reduces errors, and eases the collaboration required for software development and deployment.

2. What is the purpose of Console Output in Jenkins?

The Console Output in Jenkins provides access to the entire log for the build's execution. It includes step-by-step details of the build's progress, the commands that were executed, as well as details of any error or success messages. The Console Output in Jenkins helps developers or DevOps engineers in monitoring the build, especially if there are problems during the build.

3. What is a build in Jenkins?

A Jenkins build means an individual execution of the Job or Pipeline. Each build is assigned a unique build number and includes actions like code compilation, running tests, generating results, among others. Builds can be done manually, automatically through code changes, or even on scheduled triggers.

4. What is the role of plugins in Jenkins? Explain steps to install plugin

Plugins in Jenkins extend its functionality by adding support for tools, integrations, and additional features such as Git, Maven, Docker, and pipeline visualization. Jenkins relies on plugins to perform most automation tasks and integrate with external systems.

Steps to install a plugin in Jenkins:

1. Open the Jenkins dashboard.
2. Click on Manage Jenkins.
3. Select Manage Plugins.
4. Go to the Available tab.
5. Search for the required plugin.
6. Select the plugin and click Install without restart (or Download now and install after restart).
7. Wait for the installation to complete and restart Jenkins if required.

Conclusion

Jenkins automates CI/CD processes in DevOps, helping teams build, test, and deploy software faster and more reliably.